

SEA OF UNCERTAINTY

Advanced and Optional Rules

BOOK II - ADVANCED RULES

Development Manuscript v0.6

Operational naval warfare under conditions of incomplete information

Source baseline: public Sea of Uncertainty repository, current Prototype Rules v0.5, Design Constitution v0.3, and Design Notes.

This manuscript reorganizes source-supported rules into professional regulatory prose. Proposed or unresolved material is explicitly marked.

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ADVANCED RULE STATUS: Book II is deliberately more ambitious than the repository Core Rules. These cases are optional/provisional design modules intended for the large professional rulebook. They are not presented as already-ratified source rules. Use them by scenario agreement and revise aggressively after playtest.

35.0 ADVANCED MOVEMENT

35.1 Purpose

Advanced Movement gives the Move rating, terrain, damage, and Formation composition more influence without replacing the core Cautious/Normal/High-Tempo choice.

35.2 Movement Capacity

A Formation begins with the movement distance of its chosen Mode. Scenario or Formation data may modify that allowance by +/-1 hex. No modifier may reduce a legal Move below 1 hex unless the Formation is immobilized.

35.3 Littoral Transit

Recommended advanced treatment: entering one or more Littoral hexes during a Move adds +1 Time once per Move Action, not per hex. This keeps Littoral maneuver consequential without creating repetitive arithmetic.

35.4 Chokepoints

A scenario may designate Chokepoints. A Formation entering a Chokepoint may be required to stop, may gain Signature, or may trigger a Search/Reaction opportunity as stated by the scenario.

35.5 Damaged Propulsion

Propulsion effects modify maximum movement after Mode is chosen. A Formation whose maximum is reduced below its intended route must stop at the last legal hex.

36.0 ADVANCED SUPPORT

36.1 Support Range

Recommended baseline: a supporting Formation must be within 2 hexes of the supported Formation or within the range specified by its special capability. Air/EW Support may use larger scenario-defined ranges.

36.2 Timing

Declare Support before the supported Search, Strike, or Reaction is resolved. The supporting Formation immediately receives a Ready Time based on its Support cost even if the supported action later fails or is aborted.

36.3 Stacking

No action may receive more than +2 total Support unless a scenario explicitly permits it. Multiple Support sources may contribute, but identical Support types normally do not stack beyond +1.

36.4 Sustained Support

A scenario may allow AEW, tanker, or persistent escort Support to remain assigned. A sustained Support assignment occupies the Formation's Mission and may occupy Command Attention if centrally controlled.

37.0 ADVANCED REACTIONS

37.1 Reaction Radius

Recommended baseline: Patrol/Screen and interception Reactions operate at 1 hex. Formation data may extend this to represent long-range air or missile coverage.

37.2 Intercept

An eligible Patrol Formation may Intercept an enemy entering its Reaction Radius. The intercepting Formation may either move one hex toward the intruder or conduct a permitted Strike/Support Reaction, then becomes Reaction-spent.

37.3 Emergency EMCON

Instead of Defend/Evade, a Formation may reduce Signature immediately, forfeiting Counterattack and adding +1 Time to its next Action.

37.4 ASW Prosecution Reaction

An ASW-capable Screen that receives a submarine Contact in its protection area may conduct an ASW Search as its Reaction, subject to 48.0.

38.0 ADVANCED CONTACT AGING

Age	Advanced Effect
0 Current	No penalty
1 Usable	No penalty
2 Stale	-1 Targeting; Search may refresh normally
3 Very Stale	-2 Targeting; degrade Location one level
4+	Degrade Location each Time; Low becomes Lost

38.1 Fast Targets

Air contacts may age one additional step when they complete a high-speed mission, if the scenario uses detailed Air Operations.

38.2 Submarine Contacts

Submarine Contacts use the localization rules of 47.0/48.0 rather than this generic schedule when Advanced Submarine Operations are in use.

39.0 ADVANCED DAMAGE

39.1 Damage Effects

When a result calls for specific damage, draw or assign a Destruction Effect Card. The card is the detailed manifestation; the core Heavy/Crippled state remains the overall damage category.

39.2 Light Damage

Recommended advanced rule: Light Damage creates one temporary impairment marker. The defender chooses one currently usable capability (Search, Strike, Defense, or Move) to suffer -1 until the Formation completes Recover. Light Damage does not mark Destruction.

39.3 Crippled

Recommended advanced rule: maximum surface/sub movement becomes 1 hex; Heavy Salvo unavailable; Reaction limited to Defend or Evade; Support may not be provided unless a card specifically permits it.

39.4 Multiple Damage Effects

Specific Destruction cards stack unless impossible. If two cards would prohibit the same capability, the second card may be redrawn at scenario option to avoid null duplication.

40.0 COMPLEX ACTIONS

40.1 Definition

Under the Advanced Rules, an Action is Complex if it includes any of the following: participation in Synchronization; immediate retasking while Disrupted; High Tempo while Extended or Critical; multi-Formation Support; acting while a Broken Link is being bypassed; or a scenario-designated difficult Littoral/air/ASW operation.

40.2 Friction

A Formation with Friction adds +1 Time after completing a Complex Action unless that delay is removed by Push Through or a specific card.

41.0 HQ RECOVERY

41.1 Procedure

An HQ may conduct Recovery when no more than one of its Command Slots is currently occupied. Spend 2 Time on the HQ track or the scenario-defined command cycle to remove one Command Strain.

41.2 Restoring Slots

If Command Strain had reduced the number of available Slots, restore the lost Slot immediately when Strain falls below the threshold that caused the loss.

41.3 Command Disruption

HQ Recovery may also remove one Degraded HQ effect if the cause is no longer active, but it does not automatically restore a Broken Link to a Formation.

42.0 ADVANCED LOGISTICS

42.1 Logistics Access

A Formation has Logistics Access when it occupies or is adjacent to a friendly logistics Formation, designated replenishment hex, port, or base specified by the scenario and no rule blocks that source.

42.2 Contested Logistics

A logistics source inside an enemy Strike range or contested objective zone may still be used, but Replenishment generates +1 Signature and may trigger enemy Reaction/Search opportunities.

42.3 Fuel and Ordnance

Scenarios may split replenishment into Fuel and Ordnance. Fuel restores Endurance; Ordnance removes Weapon Expenditure. A single Replenish Action may restore one category unless enhanced logistics capability says otherwise.

43.0 REPAIR AND REORGANIZATION

43.1 Repair Classes

Treat Destruction effects as Operational, Repairable, or Scenario-Persistent. The card or scenario states the class.

43.2 Operational Reorganization

Spend Recover to remove an eligible organization-related Destruction effect such as Escort Lost when the Formation is no longer under immediate attack and the card permits reorganization.

43.3 Repairable Damage

Repairable effects require Replenishment with Logistics Access. Remove one eligible Destruction card per successful Repair/Replenish action.

43.4 Persistent Damage

Scenario-Persistent effects cannot be removed during the scenario except by a specific card or special rule.

44.0 ADVANCED COMMAND SLOTS

44.1 Release Timing

Focused Search releases its Slot after the Search resolves. Immediate Retask releases after the new Mission is accepted. Synchronization occupies its Slot until the synchronized operation resolves or is aborted. Push Through releases after scheduling the affected Formation. Restore Command releases when the restoration attempt resolves.

44.2 Sustained Tasks

A scenario may designate sustained tasks that keep a Slot occupied across multiple Ready Times, representing continuing staff supervision.

44.3 Command Range

If used, a Formation beyond its HQ command range may continue standing Missions but requires either a local Command test or Response Card for immediate retasking.

45.0 ADVANCED RESPONSE CARDS

45.1 Hand Limit

Recommended hand limit: 4 Response Cards. A player who gains a fifth must play or discard one before continuing.

45.2 Command Cycle Draw

Recommended draw: each side draws one Response Card after its HQ completes Recovery, and may also gain cards from scenario events or objectives.

45.3 Deck Construction

Scenarios may use the full 24-card Response set or build doctrine-specific subsets to create different command styles.

46.0 AIR OPERATIONS

46.1 Mission Packages

Advanced Air Operations treat aircraft as temporary mission packages launched from a base or carrier Formation rather than as permanently occupying sea hexes.

46.2 Mission Radius

Current digital baseline mission radii are 4/6/8 hexes. A scenario may assign these to Cautious/Normal/Extended air commitments or to aircraft type.

46.3 Air Mission Sequence

Launch - Transit - On Station/Strike - Recover. A mission package must have a legal recovery source when launched unless conducting a one-way scenario mission.

46.4 CAP

CAP may be assigned as Patrol/Screen Support. When an eligible Strike enters the protected area, CAP may provide Defense or Intercept under 37.0.

46.5 Sortie Turnaround

After recovery, the parent carrier/base receives the mission package's turnaround Time before that package may launch again. Friction and Destruction cards may increase turnaround.

47.0 SUBMARINE OPERATIONS

47.1 Stealth State

Advanced submarines may operate in Quiet or Exposed state. Quiet reduces Signature but restricts speed; Exposed permits High Tempo or active sensing at increased detectability.

47.2 Datum

A successful generic Search against a submarine creates a Datum rather than automatically producing full High Location. Datum represents the last reliable submarine position.

47.3 Trail

A surface/air ASW force may maintain a Trail by repeatedly refreshing the submarine Contact. Losing the Trail returns the Contact to Datum/aging procedures.

47.4 Submarine Strike

A submarine with sufficient Contact quality may conduct a torpedo or missile Strike. A torpedo Strike may reduce defender Reaction options depending on scenario range and warning.

48.0 ADVANCED ASW

48.1 ASW Information Ladder

Use Detect - Classify - Localize - Track - Attack as the advanced ASW sequence. Each successful ASW Search advances one step unless a special capability advances more.

48.2 Localization

Only Localize or Track provides targeting-quality position for a torpedo/weapon attack against a submarine.

48.3 Prosecution Time

Each ASW Search/Prosecution costs 2 Time. High-Tempo prosecution may gain +1 but causes Friction and increases the hunter's Signature.

48.4 Reacquisition

After a submarine Evades or survives attack, degrade its Contact one ASW step unless a Track is maintained by another supporting asset.

49.0 ADVANCED STRIKE RANGES AND WEAPONS

49.1 Weapon Profiles

A Weapon Profile may specify Range, Salvo Capacity, Attack modifier, target restrictions, and special defense interactions. The core Light/Standard/Heavy commitment remains the quantity decision layered over the weapon type.

49.2 Magazine Depth

Instead of one generic Weapon Expenditure marker, a scenario may give a Formation two or more Magazine boxes. Standard or Heavy Salvos consume the number stated on the profile.

49.3 Standoff Weapons

Long-range weapons may require High Location or Identified Contact quality. Some may permit launch against stale Contacts with additional targeting penalties.

50.0 LAYERED MISSILE DEFENSE

50.1 Defense Layers

Advanced missile defense may divide total Defense into Outer Air Defense, Area Defense, and Terminal Defense.

50.2 Resolution

Resolve the Strike through each eligible layer in sequence. Successful defensive effects reduce incoming Strike strength or Salvo commitment before the next layer. Any remaining attack becomes leakers and resolves against the target.

50.3 Saturation

Synchronized or Heavy attacks may degrade one or more defense layers for later attacks in the same window, replacing or refining the core -1/-2 synchronized Defense penalties.

DESIGN CAUTION: Do not use layered defense merely to add dice. It should create decisions about salvo size, attack sequencing, escorts, and conservation of defensive capacity.

51.0 ADVANCED ENDURANCE

51.1 Operational Stress

Instead of counting every three major Actions automatically, scenarios may assign Endurance Stress for High Tempo, Strike, prolonged Patrol, air sortie generation, and combat damage. Accumulating a scenario-set number of Stress marks degrades Endurance.

51.2 Fuel and Combat Endurance

High-detail scenarios may separate Fuel Endurance from Combat Endurance. Fuel primarily limits movement; Combat Endurance limits Strike, Reaction, and sortie generation.

52.0 ADVANCED DECEPTION

52.1 Deception Actions

Advanced Deception may generate False Emissions, Dummy Formations, False Logistics activity, Feints, or False Air Operations. Each creates or manipulates Contacts rather than awarding abstract deception points.

52.2 Counter-Deception

Focused Search may be declared specifically to disprove one suspected False Contact. Success removes it; failure leaves the Contact unchanged and still costs Time.

53.0 EW AND CYBER

53.1 Principle

EW and Cyber must feed existing systems: Search, Contacts, Command Slots, Synchronization, Ready Time, or Entropy. Avoid a separate resource mini-game.

53.2 Jamming

Successful Jamming normally creates Disruption and may suppress Active/Focused Search or reduce Support.

53.3 Spoofing

Spoofing creates or alters False Contacts, Contact Age, or Identity.

53.4 Cyber Command Attack

A successful command attack may cause Degraded HQ, Delayed Information, or Broken Link. Persistent effects require repeated success or a scenario-defined vulnerability.

54.0 COMMAND DOCTRINE

Doctrine	Benefit	Risk
Centralized	+1 to one synchronized operation per command cycle	Broken Link/Degraded HQ effects are harder to recover from
Mission Command	Disconnected Formations retain broader Mission/Trigger autonomy	No special synchronization bonus
Distributed	Reduced command-disruption penalties	Complex multi-Formation synchronization costs +1 additional Command attention or Time

54.1 Scenario Assignment

Command Doctrine should usually be assigned by scenario or force design rather than freely optimized by both players.

55.0 ADVANCED SCENARIO RULES

55.1 Scenario Parameters

Advanced scenarios may define weather, visibility, Rules of Engagement, neutral shipping, basing, reinforcement, logistics, cyber/EW access, political constraints, escalation thresholds, Contact setup, and Response deck composition.

55.2 Scale Overrides

A scenario may override the 20-nm/2-hour baseline if all movement, Search, Strike, aging, and mission radii are rescaled consistently.

56.0 ADVANCED VICTORY

56.1 Operational Victory Schedule

A scenario may award Victory Points for time-dependent operational conditions: safe transit, denied hexes, preserved striking power, surviving logistics, information dominance, or forced withdrawal.

56.2 Escalation and Political Cost

Some scenarios may penalize attacks on restricted targets, neutral losses, or crossing escalation thresholds even when tactically successful.

57.0 SOLO / NON-PLAYER COMMAND

57.1 Purpose

Optional solo rules may automate standing Missions and local Reactions while leaving uncertainty and card draws intact. This is a design module, not a source-derived current rule.

57.2 Behavior Priorities

Each non-player Formation receives a Mission priority list: Objective, Threat, Preservation, Support. At Ready Time, select the highest legal priority and use the prescribed posture.

57.3 Uncertainty

The automated opponent should not receive hidden player information. Use the same Contact picture available to that side when selecting targets or movement.

58.0 UMPIRED FOG OF WAR

58.1 Double-Blind Play

For professional or club play, each side may use a separate map while an umpire maintains actual Formation positions and resolves hidden Search privately.

58.2 Message Delay

The umpire may physically delay reports and orders according to Delayed Information, Broken Link, or scenario communications rules.

58.3 Contact Logs

Players should record Contact ID, last-known hex, Location, Identity, and Age to preserve a clean information history.

59.0 FORMATION CONSTRUCTION

59.1 Purpose

Formation Construction supports scenario design and later digital implementation. It is not currently a competitive point-build system.

59.2 Required Data

Every Formation design should specify type, Move, Search, Signature, Strike, Defense, ASW if applicable, Command, Endurance, weapon ranges/profiles if Advanced Weapons are used, and special capabilities.

59.3 Validation

A Formation should be tested for decision profile rather than simply point efficiency: what can it do, what does it fear, what Support does it need, and what does it sacrifice when it uses its strongest capability?

ADVANCED RULE ADOPTION: Adopt Advanced modules individually. A scenario should list exactly which Advanced sections are in force so players never have to guess which chrome applies.